

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing** *Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI-Powered Customer Support Chatbot (2024)

Designed and implemented a large-scale AI-driven chatbot leveraging retrieval-augmented generation (RAG) architecture with custom embeddings for real-time interaction. Integrated state-of-the-art large language models (LLMs) using OpenAI API and LangChain. Optimized the model deployment pipeline for efficient execution and scalability, improving response latency and reducing human agent intervention by 35%. Applied vector database management with ChromaDB to enhance knowledge retrieval and information accuracy in a distributed environment.

Tech: Python, LangChain, OpenAI API, ChromaDB, LLM Integration, RAG Architecture

Distributed Customer Sentiment Analysis Platform (2024)

Developed and optimized a sentiment classification pipeline for large-scale customer review datasets using advanced NLP techniques. Implemented scalable text preprocessing, feature extraction with TF-IDF, and leveraged logistic regression models, ensuring efficient data handling suitable for distributed training environments. Enhanced model accuracy to 89% and contributed to product improvement decisions by accurately identifying sentiment trends. The project involved collaborative optimization for model performance and deployment considerations in a multi-engineer co-design setting.

Tech: Python, TensorFlow, PyTorch, NLP, TF-IDF, Logistic Regression, Distributed Training

Telecom Customer Churn Prediction with Large-Scale Machine Learning (2024)

Engineered a predictive machine learning framework for churn prediction on high-volume telecom datasets, employing advanced data preprocessing, feature engineering, and class imbalance techniques (SMOTE). Integrated and optimized models including XGBoost within scalable distributed training environments across HPC infrastructure. Achieved an ROC-AUC of 0.87. Collaborated closely with software engineers to improve computational efficiency and hardware utilization, supporting strategic business decisions to reduce churn by 18%.

Tech: Python, Pandas, NumPy, Scikit-learn, XGBoost, SMOTE, Distributed Training, HPC

Predictive Maintenance Model for Industrial IoT Systems (2024)

Designed and trained a machine learning model utilizing Random Forest on extensive time-series sensor data from IoT devices to forecast equipment failures. Developed feature extraction techniques optimized for hardware acceleration and efficient inference on custom silicon. Achieved a 28% reduction in unexpected downtime, significantly decreasing operational costs. The project highlighted hardware/software co-design by profiling model performance on industrial-grade accelerators and suggesting training optimizations for real-time deployment.

Tech: Python, Random Forest, Time-Series Analysis, Feature Engineering, Hardware Profiling, Sensor Data

About :

Experienced AI Engineer specializing in designing, training, and optimizing large-scale AI models including Large Language Models (LLMs) and similar architectures. Proficient in distributed training strategies, model parallelism, and leveraging high-performance computing infrastructures to maximize hardware utilization and model accuracy. Skilled in collaborating within hardware/software co-design environments to deploy efficient AI workloads on custom silicon solutions. Strong programming expertise in Python and hands-on experience with ML frameworks such as TensorFlow, PyTorch, JAX, and Hugging Face Transformers. Adept at implementing advanced AI/ML techniques including sparsity, quantization, and knowledge distillation to drive training efficiency and model performance. Passionate about innovation in AI hardware/software integration, contributing to AI research and development roadmaps, and mentoring junior engineers within multidisciplinary teams.

